```
2d37849cf381: Download complete
511136ea3c5a: Download complete
e2fb46397934: Download complete
015fb409be0d: Download complete
21082221cb6e: Download complete
bbd692fe2ca1: Download complete
8db8f013bfca: Download complete
7fff0c6f0b8d: Download complete
7acf13620725: Download complete
6f114d3139e3: Download complete
7e65318a255c: Download complete
d9065eab4487: Download complete
Status: Downloaded newer image for saifikhan/cadvisor:latest
$
```

A list of images on your system can be seen, as shown below:

```
$ docker images
REPOSITORY          TAG    IMAGE ID     CREATED      VIRTUAL SIZE
saifikhan/cadvisor latest 2d37849cf381  3 months ago  17.51 MB
```

Now that we have the image, we need to run it:

```
$ sudo docker run --volume=/:/rootfs:ro --volume=/var/
run:/var/run:rw --volume=/sys:/sys:ro --volume=/var/lib/
docker/:/var/lib/docker:ro --publish=8080:8080 --detach=true
--name=cadvisor saifikhan/cadvisor:latest

root's password:
ed989c9382706ad5ac322ffedffd0f5409167ff95736b36d12af3c
adc1e06850
```

Verify that the image is running using the following commands:

```
$ docker ps
CONTAINER ID     IMAGE   COMMAND     CREATED   STATUS
PORTS           NAMES
ed989c938270   saifikhan/cadvisor:latest    "/usr/bin/cadvisor"
3 seconds ago   Up 2 seconds      0.0.0.0:8080->8080/tcp
cadvisor
$
```

Please start your browser and type in the following address:

http://localhost:8080/

You should see a page that lists all the Docker containers running on your machine. Do look at Figures 1, 2 and 3 to get an idea.

Finally, when you are satisfied, you can stop the Docker container:

```
$ docker ps
CONTAINER ID  IMAGE     COMMAND    CREATED     STATUS
PORTS        NAMES
ed989c938270     saifikhan/cadvisor:latest "/usr/
bin/cadvisor"   27 minutes ago     Up 27 minutes
0.0.0.0:8080->8080/tcp    cadvisor

$ sudo docker stop ed989c938270
root's password:
ed989c938270

$ docker ps
CONTAINER ID  IMAGE    COMMAND   CREATED           STATUS
PORTS         AMES
$
```

cAdvisor as a daemon is fairly versatile; it exports a REST API interface and supports a native storage driver for real-time time series database influxDB. When influxDB is configured to pull in the data from cAdvisor, it can perform statistical analysis of the data stream and provide rate information, e.g., context switches per second, memory page faults, etc.

cAdvisor tracks on a per-container basis the resource isolation parameters based on cgroups and namespace information, historical resource usage and network statistics. From a deployment perspective, cAdvisor can be used along with influxDB, a distributed time series database, and Grafana, an open source, feature-rich, metrics dashboard and graph editor for influxDB which works entirely from the browser environment to generate client-side graphs.

cAdvisor as an open source tool is the starting point for Docker container resource monitoring. There are further fine grained controls that can be adopted by writing scripts that manipulate the dynamic file system interface of cgroups and namespaces. In due course, many of these capabilities would either show up in the tool itself, or the ecosytem around the tool will bring in the capabilities. END

References

[1] *cAdvisor: https://github.com/google/cadvisor*
[2] *influxDB: http://influxdb.com/*
[3] *Grafana http://grafana.org/*

By: Saifi Khan

The author is passionate about open source software and believes that FOSS is one of the most enduring ways to build flexible, scalable and feature-rich infrastructure platforms. He is focused on building solutions using Docker, Microservices, CoreOS and architectural patterns for distributed computing. He can be reached at *saifi@acm.org* or *twitter.com/hiSaifi*